

Conflict and Peace

Week 7: Impact of crime and conflict on economic and institutional development

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How do conflicts affect economic and institutional development?

Conflicts impose negative impact both in short and long-run

- Short: Loss of human life, destruction of capital, increased uncertainty
- Long: Loss of tax base → Difficult to finance reconstruction, reliance on debt ↑
- It can also spill over into other regions (depress trade, increased social strains)

Conflicts undermine the foundation for institutions that support growth and development

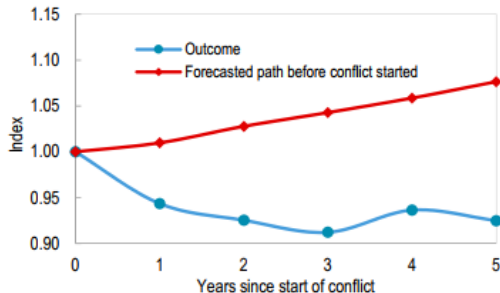
- Weakened governance: Perpetrators gain influence over decisionmakers
- Weakened accountability: General public gets less priority
- Leads to more corrupt and incapable state

These factors put many conflict-affected societies into “conflict trap”

- Conflict → Economy and institution↓ → Conflict → Economy and institution↓ →.....

Long-run effects of armed conflicts

Figure 2.19. Sub-Saharan Africa: Index of Real GDP per Capita, Actual versus Forecast

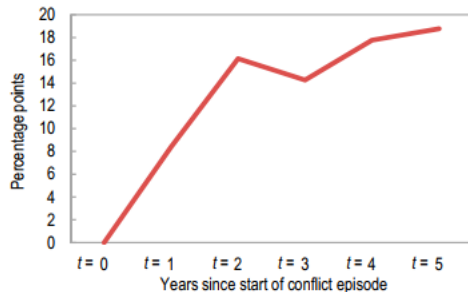


Source: IMF staff calculations.

Note: Based on 10 conflict episodes listed in Annex Table 2.9.

(a) Actual GDP vs GDP with no conflict

Figure 2.26. Sub-Saharan Africa: Cumulative Change in Debt-to-GDP Ratio during Conflict Episodes



Source: IMF staff calculations.

Note: Starting years of conflict episodes are defined based on Online Annex Table 2.9.

(b) Accrual of debt post-conflict

Can scars of conflict and crime be undone?

Wartime trauma has serious long-run development consequences

- Stunted economic development and recurrence of war if trauma is not treated
- But also difficult to escape: Many reconstruction programs do face shortfalls
 - ▶ Many reconstruction fail to foster trust across fractionalized groups and outsiders
 - ▶ Reintegration programs sometimes fail to generate interest among participants
- So what are the required approaches
 - ▶ Reshape incentives: Make it more costly to engage in conflicts
 - ▶ Winning hearts and minds: Trust across warring groups need to be rebuilt
 - ▶ For intervention led by third party: Trust in intervening entities too
- Some of these efforts do succeed in rebuilding capital and trust

Some countries do find some hope



View of Kigali business district with offices, towers and residential homes, at sunset. Stellalevi/Getty Images



Over the past 17 years, Rwanda has cleared informal settlements to make way for modern urban construction. Kigali's ambitious city master plan is expected to be fully realised by 2050. But what about the people who are pushed out in the process, and their memories? Shakirah E. Hudani presents some of their stories in her new book *Master Plans and Minor Acts: Repairing the City in Post-Genocide Rwanda*. She answers questions about Kigali's emergence from conflict, and what could be.

What has emerged from the state's vision for Kigali?

Kigali was established as a colonial outpost in 1907 by German administrator Richard Kandt. The city became the capital of an

Author



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Partners

[View all partners](#)

Question to explore today

- How does conflict hurt aggregate economic and institutional performance?
 - ▶ Destroys foundations needed for economic activities and growth
 - ▶ Distorts resource allocation to favor attackers, not general public
 - ▶ Captures state apparatus (politicians, institutions) → lead to above consequences
- How do you design effective counter-conflict programs?
 - ▶ Just addressing perpetrators may not suffice (e.g Mexico)
 - ▶ Need to reshape incentives and perceptions

We investigate how conflicts affect economic and institutional development

We then probe into how conflicts can be undone

Roadmap for today

1. Overview of the topic

2. Economic impact of conflicts

- 2.1 Pinotti (2015): The Economic Costs of Organised Crime: Evidence from Southern Italy
- 2.2 Fenizia, Saggio (2024): Organized Crime and Economic Growth: Evidence from Municipalities Infiltrated by the Mafia
- 2.3 Korovkin, Marakin (2023): Conflict and Intergroup Trade: Evidence from the 2014 Russia-Ukraine Crisis [student presentation]

3. Institutional impact of conflicts

- 3.1 Alesina, Piccolo, Pinotti (2019): Organized Crime, Violence, and Politicians
- 3.2 Brückner, Ciccone (2011): Rain and the Democratic Window of Opportunity
- 3.3 Blouin, Murkand (2019): Erasing Ethnicity? Propaganda, Nation Building, and Identity in Rwanda
- 3.4 Berrebi, Klor (2008): Are Voters Sensitive to Terrorism? Direct Evidence from Israeli Electorate [student presentation]

Pinotti (2015):
The Economic Costs of Organised Crime:
Evidence from Southern Italy

Q: Presence of organized crime $\Rightarrow$ economic development?

Organized crime is perceived as barrier to economic development

- Not just in developing countries, but also more developed countries too
- Among developed countries, Italy stands out
 - ▶ 5 regions with highest organized crime are also the poorest
- Challenge: Reverse causality among economic development and organized crime
 - ▶ We want to confirm organized crime $\uparrow \Rightarrow$ economic development $\downarrow$
 - ▶ It could be organized crime $\uparrow \Leftarrow$ economic development $\downarrow$

This paper addresses this with unexpected changes in criminal presence in select regions

- Setting: Apulia and Basilicata (normal levels of criminal presence $\rightarrow$ heavy)
- This trend arose due to changes in entry point of smuggled tobacco in Italy
- These changes are unrelated to socio-economic environments
- Generate counterfactual using synthetic control methods

History of organized crime in Italy

Description of their activities

- Supply illicit goods, extortion and threats, private protection where state is absent
- Arose from the vacuum property rights protection following 1861 Italian Unification
- Italian judicial system began to address organized crime actively since 1980s (!)
 - ▶ Before: Applied rules that treats organized crime equally with small # of individuals

Economic growth and crime in Southern Italy

- Generally, convergence between north and south stopped around 70s
- Apulia and Basilicata: High growth in 60s-70s → lowest growth rates in Italy
- The down-growth period coincides with increased organized criminal presence
 - ▶ Tobacco smuggling ↑ + Relief money after major earthquake + Criminals *in confino*

Data and empirical strategy

Measuring Crime

- Data from Parliamentary Antimafia Commission and Italian Statistical institute
- Regional GDP per capita from 1951-2005
- Value added by industry and population density accounted for

Synthetic control to create a counterfactual of the treated areas

- Idea: Actual outcome in treated area vs weighted average of control group
- Weights are chosen to match pre-treatment characteristics of treated areas
- Includes other regions except Sicily, Campania, and Calabria (pervasive OCG presence)
- Therefore, this method generates counterfactual out of the basket of control group

Trend in growth and organized criminal presence

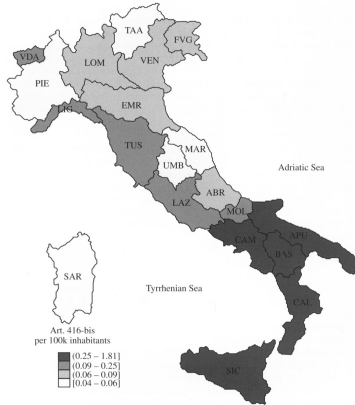


Fig. 2. Presence of Mafia-type Criminal Organisations Across Italian Regions, Average Over the Period 1983-2007

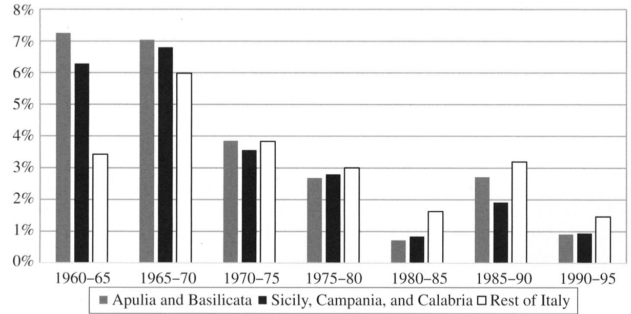


Fig. 5. Growth Rate of GDP Per Capita Across Different Areas in Italy, Different Sub-periods During the Post-war Years

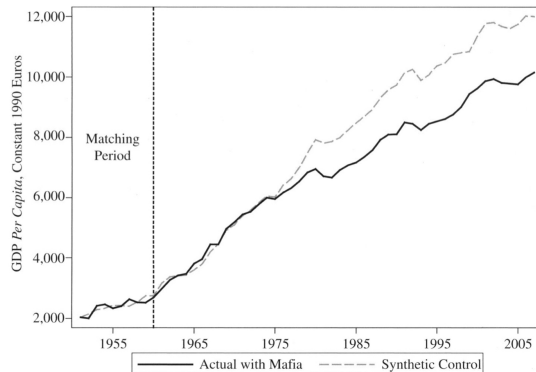
(a) OCG distribution

(b) Growth rates across time and regions

Criminal presence negatively affects growth

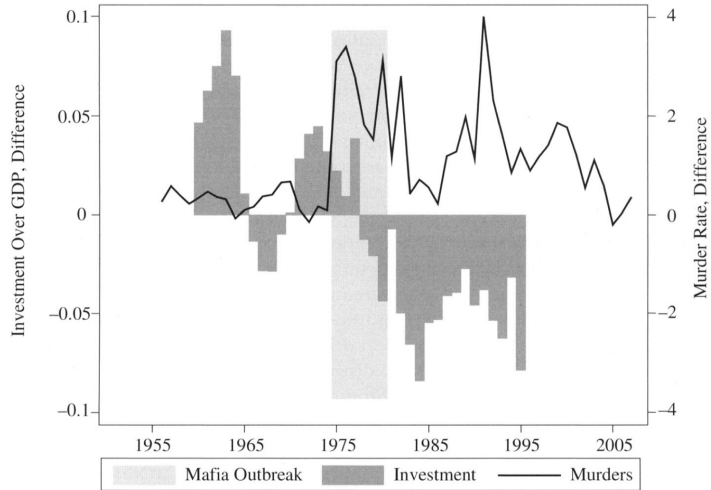
	Apulia and Basilicata	Synthetic control	Average
GDP <i>per capita</i>	2,395	2,403	3,974
Investment/GDP	0.32	0.23	0.27
VA industry	0.22	0.22	0.32
VA agriculture	0.15	0.15	0.10
VA services	0.40	0.39	0.41
VA public	0.23	0.23	0.17
Human capital	0.17	0.18	0.18
Population density (p/kmq)	134.78	105.63	145.77

(a) Synthetic control similar to treatment



(b) Actual GDP < Synthetic GDP growth

Results largely driven by decrease in economic activity



3. Private Investment and Murder Rate in the Treated Region and in the Synthetic Control, Estimated Gap, Years 1956–2007

Discussion and takeaways

Other mechanisms tested

- No shift in economic activities from formal to informal sectors
- Measured with electricity usage (even informal sectors use them)
- Electricity decreases: Suggests that the economic activities themselves decrease

Takeaways

- Presence of organized crime → Economic development ↓↓
- Driven by decrease in economic activities (investment, electricity usage)
- Macroeconomic outcomes: Does not capture effects on individual economic activities
 - ▶ Need to observe individuals as unit of analysis (last week!)
- Also, there could be evidence that informal sectors grow due to organized crime

Fenizia, Saggio (2024):
Organized Crime and Economic Growth:
Evidence from Municipalities Infiltrated by the
Mafia

Q: Break up institutions infiltrated by crime $\Rightarrow$ development?

How do you undo effects of organized crime?

- Illegal activities, extortions, hinders competition, stunt development
- What are the effects of efforts to regain control from organized crime?
 - ▶ Ultimately relates to regaining legitimacy in areas with long history of organized crime

This paper analyzes the long-run economic effects of actions to regain control

- Italy: Dismiss city councils infiltrated by the Mafia (CCD)
 - ▶ Central government dismiss mayor and city council and appoints commissioners
 - ▶ Commissioners administer with full legislative and executive authorities until next election
- Compare areas with CCDs to observationally similar municipalities w/o CCDs
- Leverage changes in economic activities: Employment, Firms, Real estate price etc

Counter-organized crime activities in Italy

Dismissal of city councils (CCDs)

- Introduced in 1991 in response to rise in Mafia influence in 1980s
- Any sign of Mafia influence detected → appoint commissioner to replace city council
 - ▶ Usually instigated by unrelated police investigations
 - ▶ Lower evidentiary bar compared to legal prosecution
- The commissioners are experienced career civil servants
- Legislative and executive authority until next local elections (roughly 24-36 months)

What do these commissioners do?

- Freeze all investment in new projects (review financial situation)
- Revoke procurement contracts, permits, licenses with connection to organized crime
- Replace municipal personnel and mandate training
- Attempt to gain trust and support of local communities (free job training etc)

Data and empirical strategy

Sources of data

- Employer-Employee matched social security data: Employment, wage, residence etc
- Real estate prices from Italian Treasury
- Politicians data from Ministry of interior (educational attainment and age)
- Councils: Ministry of interior (expenditures, revenues) & ANAC (contracts)

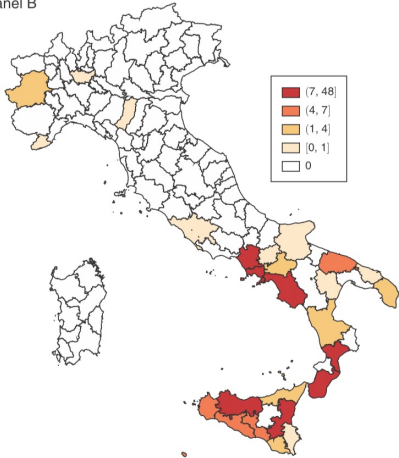
Leverage DID matching treated area to observationally similar control group (Matched DID)

$$Y_{mt} = \alpha_m + \lambda_{r(m),t} + \sum_{k=a}^b \tilde{\theta}^k I[t = t_m^* + k] + \sum_{k=a(\neq -1)}^b \theta^k I[t = t_m^* + k] \times \text{CCD}_m + u_{mt}$$

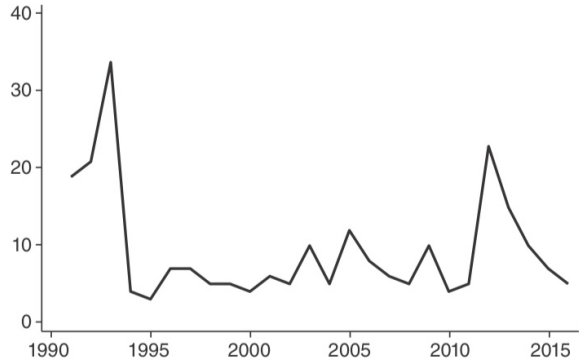
- Each treated municipalities are matched to close counterparts without CCD
- t_m^* : Year of CCD for municipality m , $I[t = t_m^* + k]$ is the event-time dummy
- $\tilde{\theta}^k$: Effect of event time dummies for control group (similar to time FE)
- θ^k is the effect of CCD k years later on the treated municipalities

CCDs over time and place

Panel B

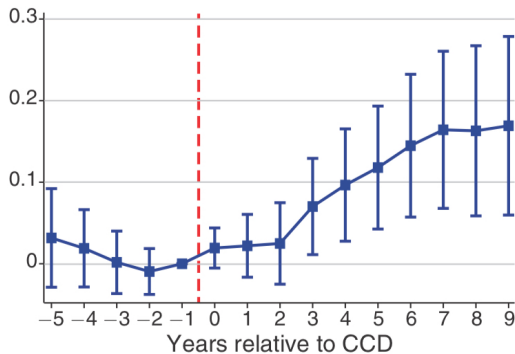


Panel A. City council dismissals due to mafia infiltration, 1991–2016

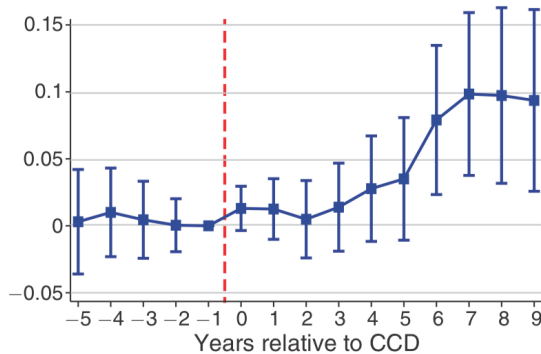


Rising employment and stock of firms

Panel A. $\log(N \text{ employees})$



Panel B. $\log(N \text{ firms})$



- Industrial real estate prices also rises (not on this slide)

Driven by weakened mafia connections

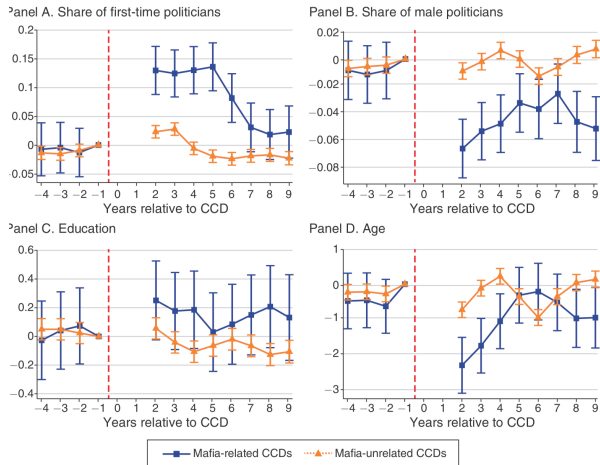


FIGURE 6. EFFECTS OF CCDs ON THE CHARACTERISTICS OF ELECTED POLITICIANS

(a) New face to local politics

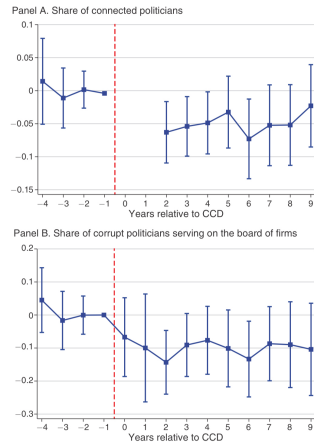


FIGURE 7. EFFECTS OF CCDs ON POLITICAL CONNECTIONS

(b) Old corrupt politicians out

Takeaways and discussions

Key takeaways

- Dismissing Mafia influence allows formal governments to reassert legitimacy
- Long term effects > Short-run effects
- New faces: Suggests that Mafia presence is actually diminishing here
- Effects of directly targeting local institutions > just illicit activities (see Mexico)

Discussions for further studies

- Institutional shake-up is required to counter long-standing crime/conflict?
- Why did the Mafia not fight back?
 - ▶ They may have changed their operation strategy (less confrontation with the state)
 - ▶ Would this work in places where organized crime actively confronts the state?

Korovkin, Marakin (2023):
Conflict and Intergroup Trade: Evidence from
the 2014 Russia-Ukraine Crisis [student
presentation]

Alesina, Piccolo, Pinotti (2019): Organized Crime, Violence, and Politicians

Q: How do organized crime influence institutional quality?

One of the worst aspects of organized crime: State capture

- Distort allocation of resources in favor of the captors (criminal organizations)
- Unjustly relax rules for their favor
- Why and how do criminal organizations take over politicians and elections?

This paper models this phenomena and identifies empirical patterns

- Violence: Signal strength and will to use violence
- Candidates scared off and influences behavior of elected politicians
- Leverages presence of Sicilian Mafia to test theoretical predictions
- Compare regions with and without significant presence of organized crime

Organized crime and politics in Italy

Key features of organized criminal group (Italy, but also others)

- Command armed men capable of exerting intimidation and influence official economy
- Sicilian Mafia colluded with certain parties
 - ▶ Christian Democrats (CD) accepted Mafia support to gain advantage against opponents
- Targeted political activists and politicians supporting left-wing parties
- Target national elections: National officials determine scope of criminal group activities

Theoretical predictions that explain criminal organization behaviors

- More likely to be violent during elections, and in swing districts (majoritarian system)
- Violence results in lower votes for honest party, higher votes for captured parties
- Anti-organized crime effort → retaliation → lower future anti-crime efforts
- These efforts from honest party decrease with rising pre-election crimes

Data and empirical strategy

Data sources

- Criminal group presence: Various NGOs (e.g Libera)
- Electoral results from Italian Interior Ministry
- Politician behavior: Transcripts of speeches from 1948-2008 in national Parliament
 - ▶ Track mentions of Mafia-related terms (almost all of them in negative light)
-

Empirical strategy leverages differences in Mafia vs non-Mafia regions

- Primarily Difference-in-differences
- But also does wide range of heterogeneity analysis (Triple differences)

$$\begin{aligned} Y_{it} = & \alpha + \beta_1 \text{Treat1}_i + \beta_2 \text{Treat2}_i + \beta_3 \text{Post}_t \\ & + \beta_4 \text{Treat1}_i \times \text{Treat2}_i + \beta_5 \text{Treat1}_i \times \text{Post}_t + \beta_6 \text{Treat2}_i \times \text{Post}_t \\ & + \beta_7 \text{Treat1}_i \times \text{Treat2}_i \times \text{Post}_t + e_{it} \end{aligned}$$

More attacks, particularly in swing states

TABLE 5
Electoral violence under different types of electoral system

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Regional data, 1887–2012			Provincial data, 1993–2004		Org. crime murder?		
	1946–92	1887–1918	1993–2004	Majoritarian elections		Yes	No	
Electoral period × Organized crime	4.303*** (0.454) [0.000]	0.901*** (0.225) [0.000]	1.610*** (0.448) [0.000]	1.512*** (0.328)	1.254*** (0.297)			
Electoral period × Organized crime × Gap	-0.372*** (0.048) [0.000]	0.004 (0.007) [0.493]	-0.007 (0.039) [0.369]					
Electoral period × Organized crime × Swing					5.686** (2.217) 0.465*** (0.173)	5.459** (2.258) 0.418** (0.170)	4.350*** (0.824) 0.019 (0.015)	1.109 (1.500) 0.398** (0.173)
means of interacted variables:								
Gap	5.211	6.024	1.356		0.196	0.196	0.196	0.196
Swing								
Observations	752	448	192	1,140	1,140	1,140	1,140	1,140
Organized crime	NO	NO	NO	NO	NO	YES	YES	YES
year FE								
R-squared	0.541	0.592	0.412	0.132	0.157	0.247	0.188	0.178

Uncooperative party vote share falls

TABLE 6
Electoral violence and electoral outcomes in Sicily, 1947–2013

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All national elections, 1948–2013			Spillovers, 1948–2013		Portella, 1947–48	
Total victims	−0.003*** (0.001)	−0.002*** (0.001)	−0.002*** (0.001)	−0.002*** (0.001)	−0.002*** (0.001)		
Political victims		−0.022*** (0.006)	−0.024*** (0.007)	−0.024*** (0.006)	−0.024*** (0.006)		
Voting for the Left, previous election			0.513*** (0.035)	0.514*** (0.035)	0.513*** (0.035)		
Total victims, spatial lag				−0.004* (0.002)	−0.067 (0.082)		
Political victims, spatial lag				−0.020* (0.010)	−0.279* (0.169)		
Distance from Portella (100s km)						0.001 (0.026)	
Elections 1948						−0.070*** (0.011)	−0.065*** (0.015)
Distance from Portella × Elections 1948						0.030** (0.012)	0.029* (0.016)
Constant	0.207*** (0.009)	0.208*** (0.009)	0.112*** (0.009)	0.112*** (0.009)	0.112*** (0.009)	0.288*** (0.034)	0.286*** (0.005)
Observations	6,533	6,533	6,171	6,171	6,171	709	709
Municipality FE	YES	YES	YES	YES	YES	NO	YES
Year FE	YES	YES	YES	YES	YES	NO	NO
Spatial lag	NO	NO	NO	Neighbours	Dist-weight	NO	NO
R-squared	0.788	0.789	0.854	0.854	0.854	0.024	0.927

Political attacks discourage politicians (esp left-wing)

TABLE 8
Electoral violence and parliamentary debates about the mafia, 1948–2008

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	MP-Legislature regression				Speech-level regression				
Total number of words (x 1,000)	0.003 (0.003)	0.003 (0.003)	0.002 (0.003)	0.115*** (0.043)	0.115*** (0.042)	0.106*** (0.037)	0.106*** (0.033)	0.094*** (0.031)	0.099*** (0.034)
Total victims	0.019*** (0.007)	0.045*** (0.013)	0.035*** (0.012)	0.469*** (0.153)	0.511*** (0.169)	0.168** (0.069)	-0.089 (0.106)	-0.034 (0.092)	-0.018 (0.084)
Political victims		-0.139*** (0.038)	-0.105** (0.041)		-0.489** (0.217)	-0.094 (0.128)	-0.045 (0.131)	-0.221* (0.130)	-0.170 (0.123)
Left			0.268 (0.205)			0.259** (0.118)	0.250*** (0.096)	-0.094 (0.325)	-0.640 (0.518)
Total victims × Left			0.056 (0.040)			0.936** (0.411)	0.920** (0.399)	0.848*** (0.326)	0.838*** (0.317)
Political victims × Left			-0.167* (0.100)			-1.092** (0.551)	-1.144** (0.513)	-0.619* (0.337)	-0.678* (0.349)
Constant	0.358*** (0.084)	0.315*** (0.085)	0.233*** (0.072)	0.163*** (0.037)	0.164*** (0.037)	0.085** (0.042)	0.664 (0.690)	1.290 (0.838)	0.428** (0.174)
Observations	655	655	655	8,833	8,833	8,833	8,833	8,833	8,833
Year FE	NO	NO	NO	NO	NO	NO	YES	YES	NO
MP FE	NO	NO	NO	NO	NO	NO	NO	YES	NO
MP x Year FE	NO	NO	NO	NO	NO	NO	NO	NO	YES
R-squared	0.009	0.027	0.045	0.009	0.009	0.020	0.046	0.027	0.007

Discussion and Takeaways

Criminal organizations and conflicts capture politicians and the states

- Strategically use elections: Timing and place
- Aim to favor the cooperating politicians by hurting the uncooperative
- Does affect behaviors of elected politicians: Reduce anti-conflict efforts
- They especially affect efforts of the honest politicians (capture ones are unaffected)

Discussions

- Generalizable? Likely (we have already seen this with insurgents in Afghanistan)
- Long run effects of such capture? (Distortionary allocations?)
 - ▶ How are public expenditures and procurement affected?
 - ▶ These could skew towards favoring captors over general public
 - ▶ We look extensively at weak states and state capture late in the semester

Brückner, Ciccone (2011):
Rain and the Democratic Window of
Opportunity

Q: Can economic shocks induce democratic transitions?

What causes movements behind democratic change?

- Many argue that economic recessions lead to democratic improvements
 - ▶ Transitory shocks → window of opportunity to contest power
 - ▶ Autocratic regime may have to make concessions
- Challenges: Democratic transition affect economics (reverse causality)

This paper isolates exogenous link between recessions and democratic transitions

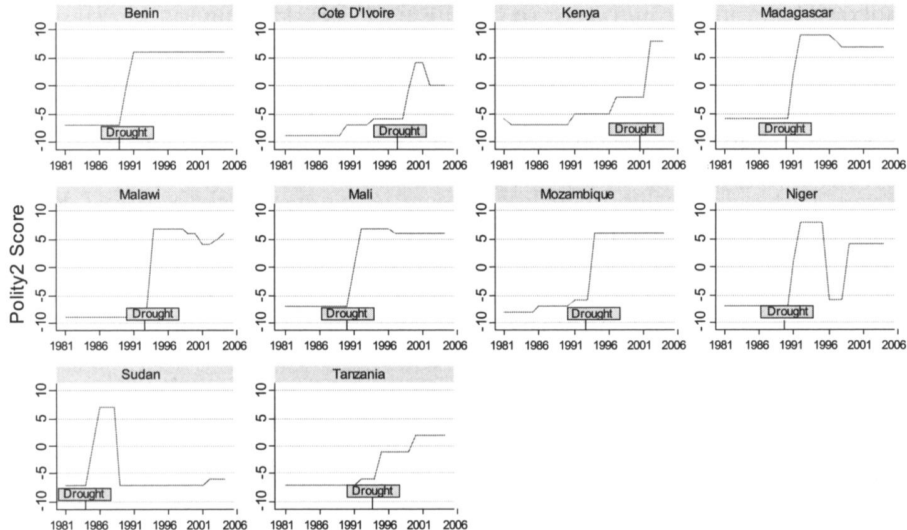
- Uses negative rainfall shocks (i.e. droughts) that affect economic performance
- Finds that even temporary shocks can lead to concessions from autocratic regimes
- Rainfall ↓↓ → GDP & opportunity cost of contesting power ↓↓ → Democratic transitions

Dataset and definitions

Sources of main measures and definitions

- Polity IV: Polity2 score that defines level of democracy (1984-2004)
 - ▶ Executive constraint + competitive executive recruitment + political participation
 - ▶ Range: -10 to 10 (0-: Autocracy, 1-6: Partial democracy, 7-10: Full democracy)
 - ▶ Dataset is available as Polity V (updated, [check here](#))
- Democratic transitions at year t : Autocracy to democracy between year t and $t + 1$
 - ▶ Vice versa for autocratic transitions at year t
- Democratic step at year t : Becomes partial/full democracy between year t and $t + 1$
- Rainfall estimates: NASA Global Precipitation Climatology Project

Democratic transitions in sample countries



Empirical strategy

How to isolate exogenous changes to economic conditions

- Rainfall shock: Affect income through agricultural productivity
- When income is accounted for, rainfall $\nRightarrow$ democratic transition on its own
- These constitute relevancy and exogeneity conditions for instrumental variables

Regression specification

$$\Delta D_{ct} = \alpha_c + \phi_c \times t + \delta_t + \beta_0 \log y_{ct} + \beta_1 \widehat{\log y_{ct-1}} + \epsilon_{ct}$$

with the first stage

$$\log y_{ct-1} = a_c + f_c \times (t - 1) + d_{t-1} + b \log \text{rainfall}_{ct-1} + e_{ct-1}$$

- Why $t - 1$: Time it takes from shock to action

Democratic transitions following negative rainfall shocks

TABLE V
INCOME SHOCKS AND POLITY CHANGE^a

	$\Delta\text{Polity2}$				$\Delta\text{Exconst}$	$\Delta\text{Polcomp}$	ΔExrec
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	2SLS	2SLS	LS	LS	2SLS	2SLS	2SLS
Log GDP, $t - 1$	-18.021** [0.049]	-21.410** [0.026]	-0.045 (0.348)	-0.836 (0.564)	-5.809* [0.073]	-7.680** [0.037]	-6.137* [0.054]
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	955	902	3191	955	902	902	902
First Stage for Log GDP per capita, $t - 1$							
Log rainfall, $t - 1$	0.079*** (0.029)	0.077*** (0.029)			0.077*** (0.029)	0.077*** (0.029)	0.077*** (0.029)
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	955	902	3191	955	902	902	902

Democratic transitions following negative rainfall shocks

TABLE IX
INCOME SHOCKS AND TRANSITIONS TO DEMOCRACY^a

	Democratic Transition			Democratization Step		
	(1) LS	(2) 2SLS	(3) 2SLS	(4) LS	(5) 2SLS	(6) 2SLS
Log GDP, $t - 1$	0.056 (0.058)	-1.285** [0.027]		-0.053 (0.051)	-1.471** [0.029]	
Country specific recession, $t - 1$			0.235** [0.027]			0.279** [0.029]
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	700	700	700	867	867	867
First Stage for Log GDP per capita/Country Specific Recession, $t - 1$						
Log rainfall, $t - 1$		0.095*** (0.037)	-0.519*** (0.164)		0.094*** (0.032)	-0.494*** (0.151)
Country fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Country time trend	Yes	Yes	Yes	Yes	Yes	Yes
Common time effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	700	700	700	867	867	867

Takeaways and discussions

Economic shocks induced by (lack of) rainfall → democratic transition

- Suggests that transitory shocks open up opportunities for change (protests)
- These apply to countries reliant on agriculture
 - ▶ In countries with low agriculture, these effects do not appear

Discussions

- Connects to how economic conditions shape incentives to contest power
- Also show that these changes can lead to structural and institutional changes
- Can this be extended to other exogenous sources to economic shocks?
 - ▶ Sudden disruption of trade opportunities
 - ▶ Resource shocks from foreign countries (esp. for countries importing resources)

Blouin, Murkand (2019):
Erasing Ethnicity? Propaganda, Nation Building,
and Identity in Rwanda

Q: Can you undo ethnic divisions that cause conflicts?

How do you bridge divides in conflict-ridden society?

- Conflicts, institutional fragility, low growth more likely in countries with ethnic divide
- Need to increase ethnic trust, cooperation & reduce salience of ethnics
 - ▶ Conversely, nation-building that addresses these points may be effective

This paper examines how propaganda can be used in nation-building

- Propaganda: Govt-owned Radio Rwanda messages on reconciliation
 - ▶ Contrast to hate message spread by RTLM (Yanagizawa-Drott 2014)
- Contention as to whether inter-ethnic reconciliation brought real progress

Mr. Kagame has created a nation that is orderly but repressive ... Against this backdrop, it is difficult to gauge sentiment about the effectiveness of reconciliation efforts (New York Times, 2017)
- Leverage geographical traits that determines the broadcast reception

Ethnicity and post-civil war nation-building in Rwanda

Post-war Rwanda

- Three ethnic groups: Hutu, Tutsis (both 97%), and Twa
- Tutsi superiority over Hutus during colonial period → inter-ethnic resentment ↑
- 1994 Genocide: 70% of Tutsis slaughtered
- Post-conflict: RPF (Tutsi-controlled) asserted control → Paul Kagame took presidency

Reconciliation under Kagame Administration

- Promote new Rwandan identity: Media, textbooks, and enforced social interaction
- *Umuganda*: Get individuals from all ethnicities for a collective project on public goods
- Messages were spread using radio (Radio Rwanda)

Data and empirical strategy

Data sources (lab-in-the-field)

- Salience of Identity Test (SIT) score from participants
- Games to identify inter-ethnicity cooperation: Partner selection game and trust game
- Individual data: Demographics (not ethnicity - illegal to ask), socioeconomic status
- Radio signal data from various geospatial sources

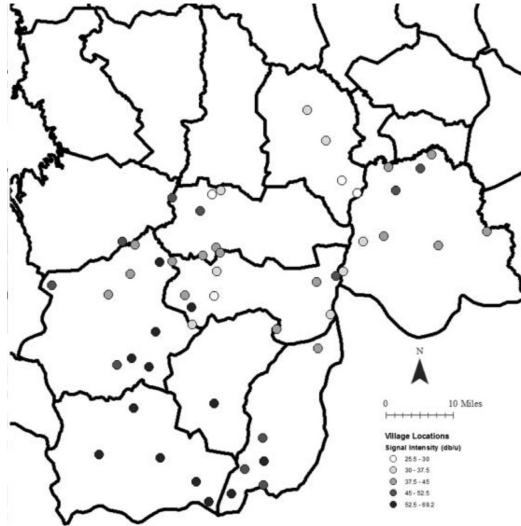
Empirical strategy uses randomization and variation across areas

- Leverages differences in radio reception across 27 transmission towers

$$Y_{ivd} = \alpha_0 + \phi_d + \beta \text{Radio}_{vd} + \Gamma X_{ivd} + e_{ivd}$$

- Radio_{vd} : 1 if signal strength $> 45dB$ at village v in district d
- Outcome at individual level

Variation in radio strength across locations



Lower salience following exposure to radio

TABLE 3
SIT SCORES AND RADIO RWANDA RECEPTION

	DEPENDENT VARIABLE				
	SIT Score (Within Error/Total Error)			SIT Score None (Within Error ² / Total Error)	SIT (Within Error/Total Error > .42)
	(1)	(2)	(3)	(4)	(5)
Radio Rwanda	-.133*** (.0463)	-.0960*** (.0289)	-.106** (.0464)	-.319** (.132)	-.167*** (.0614)
Controls	Baseline	Baseline	Full	Full	Full
Mistake fixed effects	No	Yes	Yes	Yes	Yes
Observations	438	438	438	438	438
<i>R</i> ²	.213	.402	.426	.416	.285
Control group mean of dependent variable	.855	.855	.855	1.243	.916

More inter-ethnic trust and cooperation

	# Other Ethnicity Chosen/Min{Total, 5}	
	(1)	(2)
Radio Rwanda	.156*** (.0274)	.171*** (.0360)
Controls	Baseline	Full
Observations	438	438
R^2	.406	.421
Control group mean of dependent variable	.420	.420

(a) Partnership game

	Out-Group Trust (1)	In-Group Trust (2)
Radio Rwanda reception	.250** (.105)	.0462 (.170)
Equality of coefficients (p -value)	.068	
Controls	Baseline	
Observations	438	438
R^2	.159	.127
Control group mean of dependent variable	2.92	3.17

(b) Trust game

Takeaways and discussions

Key takeaways

- Reconciliation program made some significant progress
 - ▶ Salience of ethnicity ↓, Trust across ethnic groups ↑
- Nation-building that address inter-ethnic relationship may be productive

Discussions for further studies

- Is this effect long-lasting? Unsure
 - ▶ Would need a follow-up investigation
- What happens if the focus of propaganda shifts? Unsure
 - ▶ Current reconciliation is the personal agenda of President Kagame
 - ▶ What happens if priority shifts?
- Generalizable outside of autocratic (i.e. lack of media freedom) settings? Unsure

Berrebi, Klor (2008):
Are Voters Sensitive to Terrorism? Direct
Evidence from Israeli Electorate [student
presentation]

Takeaways and remaining questions

Takeaways

- Conflicts can hurt economic and institutional development
 - ▶ It could shift to illicit sectors (i.e. increase undetectable activities)
 - ▶ But the volume of economic activities can decrease
 - ▶ State apparatus captured to bend to will of the insurgents/organized criminals
- However, these can be reversed with right actions
 - ▶ Shake-up institutional factors that cause conflicts (intergroup distrust and incentives)
 - ▶ Reduced costs of protest can lead to institutional progress

Remaining questions

- How do individuals respond to or mitigate these effects?
 - ▶ Some choose to protest
 - ▶ Others migrate to seek new opportunities
- Which types of institutions are vulnerable or resilient to these damages?
 - ▶ Need to decompose state capacity and characterize “weak states”